

Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



● SafeOT-typed (ours)

■ FDPI (constraint-based)

◆ SAC-Lagrangian (penalty-based)

▲ SAC + fixed penalty (penalty-based)

▼ SDAC (multi-constraint SOTA)

+ SRCPO (multi-constraint SOTA)

— typed Pareto frontier (soft ≤ 25)

⋯ SAC-Pen coefficient sweep

--- FDPI p_f sweep

--- SAC-Lag cost-limit sweep